Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building **robotics-adjacent**, **embedded-style**, and **AI-enabled** systems using **C++**, **Python**, and **Linux-based** workflows, with hands-on strengths in simulation, debugging, and cross-disciplinary integration.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **Python-based simulation pipelines** for autonomous systems, processing **100K+ trajectory logs** to evaluate long-horizon control behavior and improve planning performance by **18%**.
- Produced **validation documentation**, **debug logs**, and reproducible experiment traces to accelerate issue triage across software, model, and system boundaries.

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Developed robust backend components on **Azure SQL/PostgreSQL** for 500+ users, reducing latency from **1.2s to 350ms** through schema design and query tuning for real-time decision workflows.
- Implemented fault-aware **validation and release pipelines** for 5K+ monthly jobs with reconciliation checks, improving **runtime reliability** and system maintainability.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered cross-platform API/ETL services with **OOP**, **AWS Glue**, and **S3**, improving response times by **30%** and pipeline efficiency by **68%** in production environments.
- Maintained 20+ **unit/integration suites** using Jenkins, PyTest, and Mockito across **3 operating systems**, strengthening regression coverage and integration confidence.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Built production features across a **Flask + React** platform handling **10K+ daily requests**, integrating modular APIs and operationally safe data ingestion paths.
- Added concurrent execution and **CI/CD controls** via GitHub Actions, reducing turnaround time by **60%** while preserving stable behavior under peak load.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Implemented **REST API** and **WebSocket** modules supporting 8+ workflows and 2M+ synchronized records with structured logging for production diagnostics.
- Automated deployment/testing through **Jenkins + Docker**, reducing deployment time by **50%** and improving release consistency across environments.

Skills

Languages: **C++**, **Python**, Java, JavaScript/TypeScript, SQL
Robotics & AI: **Autonomous systems**, simulation pipelines, perception-model integration, runtime validation, sensor-data workflow familiarity
Systems: **Embedded Linux familiarity**, cross-platform debugging, performance tuning, integration testing, release hardening
Backend & Tooling: Flask, FastAPI, REST APIs, PostgreSQL, MySQL, AWS (S3, Glue), Jenkins, GitHub Actions, Docker, Kubernetes, Git
Engineering Practices: Agile/Scrum, code reviews, technical documentation, issue triage, multi-disciplinary collaboration

Projects

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented **6 scheduling algorithms** with interactive controls and visual execution traces across **1K+ process combinations**, strengthening systems debugging and performance analysis.

CaptionCraft – Image Caption Generator Live | *Transformers, PyTorch, Flask*
- Benchmarked BLIP/GIT/ViT-GPT-2 over a **10K image / 100K caption** dataset, improving caption accuracy by **15%** with reproducible evaluation and inference workflows.

HMIS Live | *MERN, ABDM APIs, Twilio*
- Designed API-driven operational workflows with monitoring-backed controls supporting **100+ daily appointments** and **2x faster** record retrieval under production use.